

10	(a) e.g. beam is divergent/obeys inverse square law absorption (in block) scattering (of beam in block) reflection (at boundaries) <i>(any two sensible suggestions, 1 each)</i>	B2	[2]
(b)	(i) $I = I_0 \exp(-\mu x)$ $I_0/I = \exp(0.27 \times 2.4)$ $= 1.9$	C1	
		A1	[2]
	(ii) $I_0/I = \exp(0.27 \times 1.3) \times \exp(3.0 \times 1.1)$ $= 1.42 \times 27.1$ $= 38.5$	C1	
		A1	[2]
(c)	<i>either</i> much greater absorption in bone than in soft tissue <i>or</i> I_0/I much greater for bone than soft tissue	B1	[1]